When Trading Dries Up

A liquidity signal for a long-only fund

Group 7
Silver Fund

Built on 2011–2018, tested on held-back 2019–2024

The idea

- Most investors avoid illiquid stocks
- When a stock's trading collapses, it must pass to a few patient buyers
- That hand-off is slow, so the price **drifts down for months**
- Shorting these stocks is costly, so nobody trades the drift away

What is a liquidity event?

Illiquidity (Amihud): price move per dollar traded, last 21 days

$$\mathrm{ILLIQ}_{i,t} = \frac{1}{21} \sum_{d=t-20}^{t} \frac{|r_{i,d}|}{\text{dollar volume}_{i,d}}$$

Shock: against the stock's own past year

$$x_{i,t} = \log \mathrm{ILLIQ}_{i,t} - \frac{1}{12} \sum_{k=1}^{12} \log \mathrm{ILLIQ}_{i,t-21k}$$

Remove the fund's six scores $f_{i,t}$, then standardize across stocks

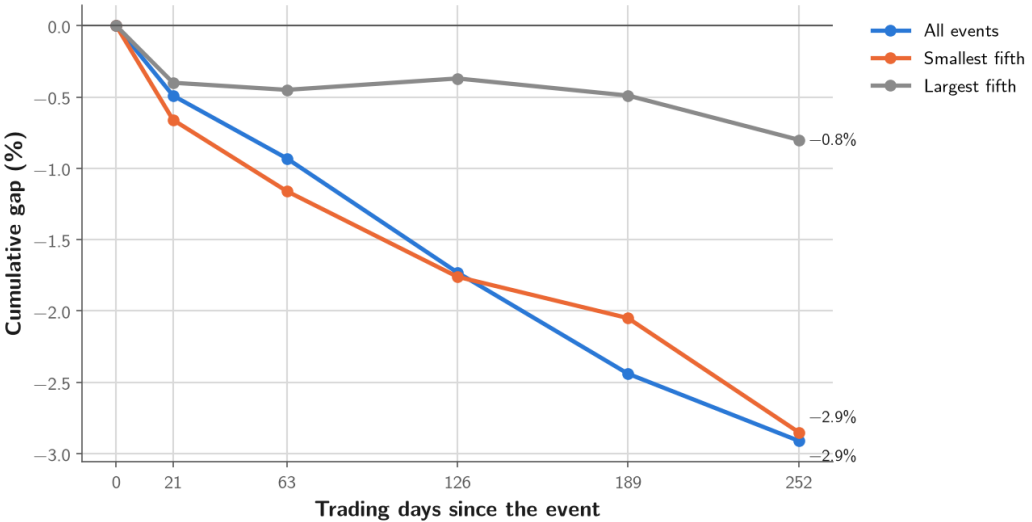
$$x_{i,t} = b_t^\top f_{i,t} + e_{i,t}, \qquad z_{i,t} = \frac{e_{i,t} - \bar{e}_t}{\mathrm{sd}_t(e)}$$

Event: $z_{i,t}$ in the day's top 5%, with no flag in the previous 21 trading days

Bottom line

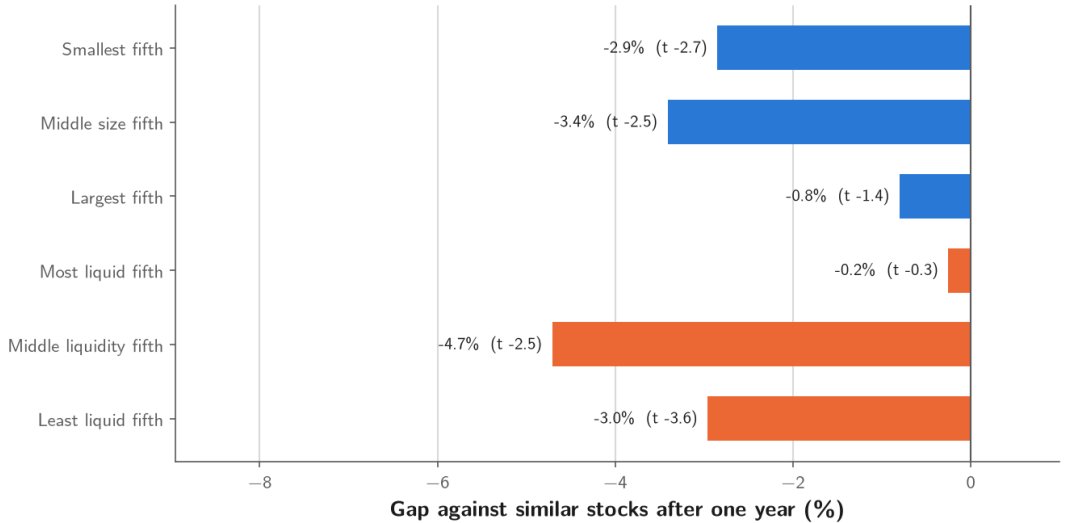
- **Real:** flagged stocks lag similar stocks by about 3% over a year
- **Just avoiding them:** right direction, too noisy
- **Optimizer, long-only after costs:** +0.17% a year, IR 0.56, on held-back 2019–24
- **Small:** one input to the fund's alpha

After trading dries up, the stock keeps falling



Flagged stocks minus similar stocks (same size, same liquidity before the event), 2006–2018.

Mostly in small, illiquid stocks



One year after the event, 2006–2018. Blue: by size. Orange: by how illiquid the stock already was.

It is not something we already know

	How much less flagged stocks earn, %/month (t)
Had a liquidity event in the last month	0.18 (2.5)
... also controlling for last month's return	0.17 (2.4)

Positive = flagged stocks earn less. Monthly regressions, 2008–2018, controlling for size, prior illiquidity, momentum and long-term reversal.

Just avoiding them points the right way, but is mostly noise

2011–2018	Weight moved	Return %/yr	Stock-specific	Factor tilts	Risk	IR (t)
Just avoid them, daily	8.6%	+0.05	+0.09	−0.04	0.47%	0.11 (0.3)
Just avoid them, monthly	8.7%	+0.10	+0.09	+0.01	0.43%	0.23 (0.7)
Optimizer, long-only + costs, daily	—	+0.12	—	—	0.11%	1.04 (2.9)

- Stock-specific part is positive: the flagged stocks did lag
- Reweighting 8–9% of the index adds factor tilts: about half the risk, about no return
- The optimizer hedges the tilts and sizes each bet by expected drift

How we bet on it

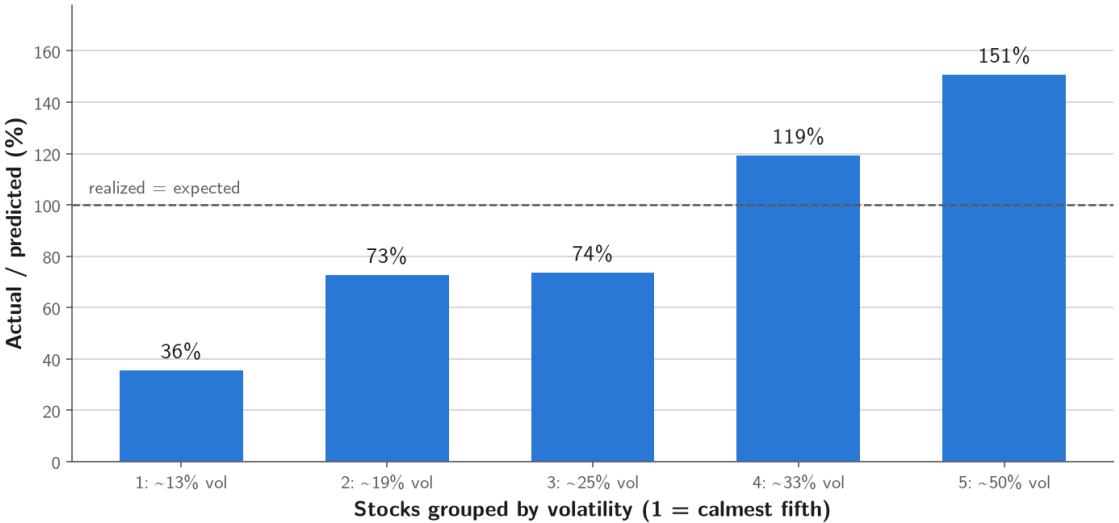
Our alpha (Grinold–Kahn form): $\alpha_i = \sigma_i \times \bar{d}_{\text{age of event}}$

- σ_i: volatility $\bar{d}$: past drift vs similar stocks, per unit of volatility no recent event: $\alpha = 0$

Optimizer: maximize $h^\top \alpha - \frac{\gamma}{2} h^\top V h - \lambda \cdot$ spread costs, active risk $\leq 1\%$

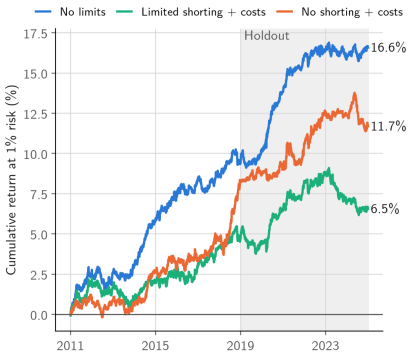
Constraint	No limits	Limited shorting + costs	No shorting + costs
Fully invested: active weights sum to zero	✓	✓	✓
Zero beta to the benchmark	✓	✓	✓
Net zero within each size × liquidity group	✓	✓	✓
Short at most 0.25% per stock ($b + h \geq -0.25\%$)	–	✓	–
Long-only: no weight below zero ($b + h \geq 0$)	–	–	✓
Pays half the bid–ask spread on every trade	–	✓	✓

Calm stocks fell less than we expected, volatile stocks more



Actual drift as a share of what we predicted, by volatility group (earlier version, 2011–2018).

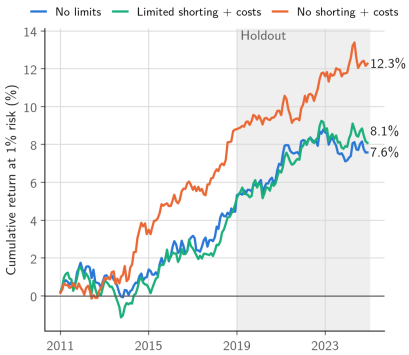
Daily rebalancing, 2011–2018 and holdout 2019–2024



	No limits		Limited shorting + costs		No shorting + costs	
	2011–18	2019–24	2011–18	2019–24	2011–18	2019–24
Active return, %/yr	+1.30	+1.44	+0.57	+0.35	+0.12	+0.17
Active risk (realized)	1.11%	1.19%	0.97%	1.14%	0.11%	0.29%
IR (t)	1.17 (3.3)	1.20 (3.0)	0.59 (1.7)	0.31 (0.8)	1.04 (2.9)	0.56 (1.4)
Max drawdown	−1.53%	−1.37%	−1.79%	−3.35%	−0.16%	−0.70%
Positive years	8/8	5/6	6/8	3/6	7/8	4/6
Turnover (one-way), ×/yr	17.5	16.1	5.8	5.5	0.4	0.6
TC vs no limits	ref.	ref.	0.84	0.82	0.33	0.36
Spread cost, %/yr	2.64	3.04	0.52	0.48	0.01	0.02

Lines scaled to equal risk so their shapes compare; the table has the real numbers. Long-only runs at only 0.1–0.3% risk: the chart scales it up about 5–10×, which it can't do in practice.

Monthly rebalancing, 2011–2018 and holdout 2019–2024



	No limits		Limited shorting + costs		No shorting + costs	
	2011–18	2019–24	2011–18	2019–24	2011–18	2019–24
Active return, %/yr	+0.62	+0.67	+0.43	+0.54	+0.14	+0.18
Active risk (realized)	1.11%	1.28%	0.71%	0.99%	0.12%	0.30%
IR (t)	0.56 (1.6)	0.52 (1.3)	0.60 (1.7)	0.54 (1.3)	1.10 (3.1)	0.58 (1.4)
Max drawdown	−2.03%	−2.15%	−1.89%	−1.46%	−0.10%	−0.43%
Positive years	5/8	5/6	5/8	5/6	8/8	5/6
Turnover (one-way), ×/yr	5.7	5.3	1.4	1.5	0.1	0.3
TC vs no limits	ref.	ref.	0.58	0.66	0.24	0.32
Spread cost, %/yr	1.04	1.21	0.13	0.10	0.01	0.01

Lines scaled to equal risk so their shapes compare; the table has the real numbers. Long-only runs at only 0.1–0.3% risk: the chart scales it up about 5–10×, which it can't do in practice.

How much to trust it

- Built on 2011–18, frozen, then run on 2019–24 **once**
- Long-only after costs, 2019–24: $+0.17\%$ a year, IR 0.56
- Six years is short: true IR likely between -0.3 and 1.4

Conclusion

- When trading collapses, the stock lags for months
- A long-only fund can capture it, with the optimizer
- It is small: one input to the fund's alpha
- Next: combine it with the fund's six scores

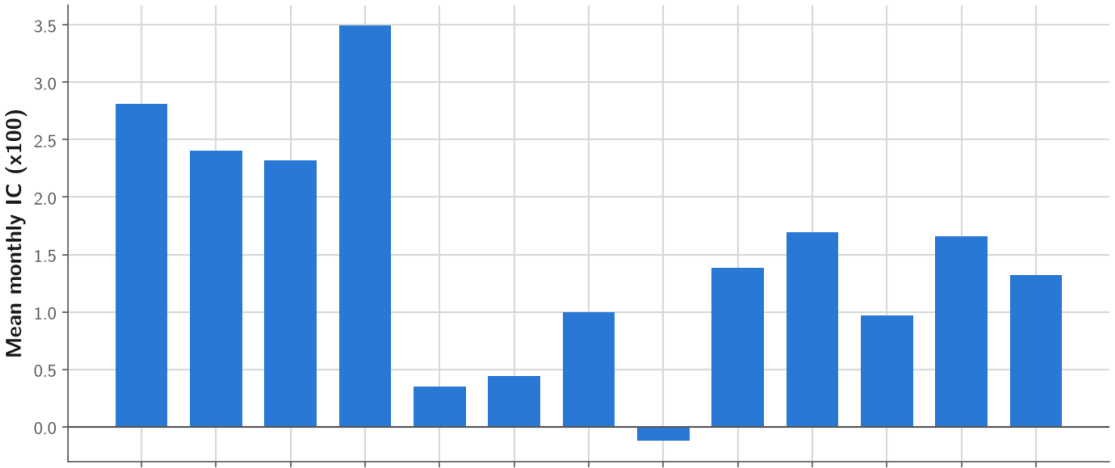
Appendix

Appendix: why does long-only have the higher IR?

Daily		Risk	Return before costs, %/yr	Spreads paid	Return after costs, %/yr	IR before costs	IR after costs	Turnover (one-way)
2011–18	Limited shorting	0.97%	+1.09	0.52	+0.57	1.12	0.59	5.8×
	No shorting	0.11%	+0.13	0.01	+0.12	1.16	1.04	0.4×
2019–24	Limited shorting	1.14%	+0.83	0.48	+0.35	0.73	0.31	5.5×
	No shorting	0.29%	+0.19	0.02	+0.17	0.64	0.56	0.6×

- Before costs, the two have about the same IR
- Limited shorting trades about 6× a year, mostly in small stocks: spreads take about half its return
- Long-only trades about 0.5× a year in larger stocks: spreads cost almost nothing
- Long-only takes only 0.1–0.3% risk: higher IR after costs, but less return

Appendix: new information beyond the fund's scores? IC by year



IC: how well our score ranks next month's stock-specific returns, after removing the fund's six scores. Positive in 12 of 13 years.

Appendix: what did we change, and when?

Daily runs, %/yr (IR), no costs	No limits	No shorting
1. Flag that turns over daily	−0.9 (−0.15)	—
2. Event clock	−2.77 (−0.49)	−0.12 (−0.24)
3. + no positive alpha, hedge with similar stocks	+3.23 (0.58)	+0.36 (1.48)
4. + alpha scaled by volatility	+6.84 (1.17)	+0.19 (1.82)

- v2: each event gets an age v3: no positive alpha; hedge with similar stocks v4: alpha scales with volatility

Appendix: how we picked the short limit (2011–2018)

Short limit per stock	Active risk (model)	IR	Turnover (one-way), ×/yr	Monthly IR
0.05%	0.47%	1.14	8	0.74
0.1%	0.85%	1.02	13	0.65
0.25%	**1.68%**	**0.97**	**27**	**0.54**
0.5%	2.65%	0.94	45	0.54
1%	3.90%	1.03	70	0.48
2%	4.71%	1.12	88	0.50

IRs are within noise of each other, so we picked the limit by risk (bold). Earlier design, 5% risk.

Appendix: every portfolio with and without trading costs (2011–2018)

		Return, %/yr	IR (t)	Risk	Turnover (one-way), ×/yr	Spread cost, %/yr
Daily rebalancing						
No limits	no costs	+1.30	1.17 (3.3)	1.11%	17.5	2.64*
No limits	with costs	+0.74	0.71 (2.0)	1.05%	6.9	0.58
Limited shorting (0.25%)	no costs	+1.09	1.07 (3.0)	1.02%	15.2	2.34*
Limited shorting (0.25%)	with costs	+0.57	0.59 (1.7)	0.97%	5.8	0.52
No shorting	no costs	+0.16	1.97 (5.6)	0.08%	0.9	0.08*
No shorting	with costs	+0.12	1.04 (2.9)	0.11%	0.4	0.01
Monthly rebalancing						
No limits	no costs	+0.62	0.56 (1.6)	1.11%	5.7	1.04*
No limits	with costs	+0.38	0.51 (1.4)	0.76%	1.6	0.13
Limited shorting (0.25%)	no costs	+0.53	0.52 (1.5)	1.02%	4.7	0.89*
Limited shorting (0.25%)	with costs	+0.43	0.60 (1.7)	0.71%	1.4	0.13
No shorting	no costs	+0.07	0.98 (2.8)	0.07%	0.3	0.02*
No shorting	with costs	+0.14	1.10 (3.1)	0.12%	0.1	0.01

* = spread cost if it were traded, not charged.

Appendix: where does the return come from? (2011–2018)

Return, %/yr	Event 1–21 days ago	Event 22–504 days ago	No recent event
Daily, no limits	+14.88	−3.43	−4.64
Daily, no shorting	+0.12	+0.08	−0.01
Monthly, no limits	+8.02	+0.09	−4.09
Monthly, no shorting	+0.09	+0.06	−0.05

Return by the age of the event behind each position.